KAPLAN INTERNATIONAL PATHWAYS

Assessment Task Information

Key details:	
Assessment title:	**Practical Assignment: Programming tasks**
Module Name:	Computer Architecture
Module Code:	IY497
Teacher's Name:	Fakiha Akbar
Assessment will be set on:	26/05/2025
Feedback opportunities:	Peer feedback in class, draft submission 15th June, 2025 on VLE
Assessment is due on:	30/06/2025 (Monday) before 9:00AM
Assessment weighting:	50%

Assessment Instructions

What do you need to do for this assessment?

Task:
Answer Part 1 (compulsory) and then any 2 questions from Part 2.

This is an assessment on assembly programming; therefore, your answers must be assembly source codes along with comments where appropriate and required.

Your source code must be reasonably indented, commented and readable.

Part 1 – You must answer this question (Compulsory)

Implement an assembly fragment, given the values of R and b are in $s0 and $s1 to implement the following function and saving result in new location.

$$A = \frac{\pi * R^2 * b}{2}$$

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Program 5 Class Solution

Part 2 – Answer any 2 questions only

1. At an international airport, **baggage tracking records** are loaded into memory. Each luggage record consists of the following structure:
   The first ten bytes are **Luggage Tag ID,** stored as ASCII
   The next byte is the **priority score** (Integer value (0–255) where higher = higher priority to process
   The next 2 bytes are the **Terminal zone code**

   There are 55 such luggage records stored sequentially in memory. The memory begins at address $10000. The final record is identified by a Zone Code of $FFFF.

   Write an assembler subroutine "FindMaxPriority" that will traverse the entire luggage records in the memory and find the highest priority score. If there are more than one luggage records with similar highest priority score, the first luggage in the list with the highest priority score is returned. The subroutine should return and store the starting address of this record in address register a5.

2. You have been tasked to develop an encryption algorithm that involves prime factorial numbers. Develop a subroutine called “Primer” that will accept a number as an input and will produce and store a maximum of first five prime factorials of the input in a1 to a5 registers.

   Use the total decimal value of the ASCII representation of **your last name** as the input and demonstrate your implementation (using an input and output table) in detail.

   Ex: last name is BOND

Alphabet	ASCII Decimal
B	66
O	79
N	78
D	68
Total	291

3. Write an assembly program that will solve a 4-term polynomial. The polynomial you choose must satisfy all the following requirements:

   a. There must be at least one exponentiated term.
   b. There must be at least 3 different variables, which will be derived from the letters of **your last name** (e.g., if your last name is **BOND**, the variables can be B, N, and D). If your last name does not contain 3 unique characters, you may introduce additional variables of your choice.
   c. One of the terms is a multiplication of any of the two variables.

Guidance:

For this assessment you should make use of the following formative activities that you have already completed. These activities have been designed to support this summative assessment:

- Week 11 – 20 formative tasks (VLE)

Please note:

This is an individual assessment so you should not work with any other student.
You may request the tutor to arrange a time for you to seek clarity and feedback on programming tasks.

Structure:

Your submission should be formed of one document containing the programming codes. Each answer must be clearly numbered.

Theory and/or task resources required for the assessment:

You must refer to your module content for this assessment. You may find additional support from the resources listed in your module handbook.

Referencing style:

NOT required

Expected word count:

Not Applicable

Learning Outcomes Assessed:

- Demonstrate the ability to read and write simple programs in assembly language.

Submission Requirements:

You must include the following paragraph on your title page:

> *I confirm that this assignment is my own work.*
> *Where I have referred to academic sources, I/we have provided in-text citations and included the sources in the final reference list.*

How to avoid academic misconduct

You should follow academic conventions and regulations when completing your assessed work. If there is evidence that you have done any of the following, whether intentionally or not, you risk getting a zero mark:

Plagiarism & poor scholarship

- stealing ideas or work from another person (experts or students)
- using quotations from sources without paraphrasing and using citations

Collusion

- working together with someone else on an individual assessment, e.g., your work is corrected, rephrased or added to by another (both parties would be guilty)

Buying or commissioning work

- submitting work as your own that someone else produced (whether you paid for it or not)

Cheating

- copying the work of another student
- using resources or aids that are not permitted for the assessment

Fabrication

- submitting work, e.g., laboratory work, which is partly or completely made up. This includes claiming that work was done by yourself alone when it was actually done by a group

Personation

- claiming to be another student and taking an assessment instead of them (both parties guilty)

Specific formatting instructions:

You must type your assessment in Arial font 11, with single spacing.

You must submit the assessment electronically via the VLE module page. Please ensure you submit it via Turnitin.

Assessments submitted after the submission deadline may incur penalties or may not be accepted.

Addition submission information – check you have done the following:	
Formatting	Consistent font, spacing, page numbers, formatting and subheadings
Citations	Correct format and location throughout the report
Referencing	Harvard referencing system used correctly in the reference list
Summarising	Summarising the results of research
Paraphrasing	Paraphrasing the contents of research findings
Spell check	Spell check the report
Proof-reading	Proof-reading completed
Grammar	*Grammarly* has been used to check the report

How will this assessment be marked?

The assessment will be marked using the following areas and weightings:

- Part 1 (30%) (Compulsory)
  - Program presentation (5%)
    - Detailed description of the data types and data structures used

        - Source code reasonably indented, commented and readable
    - Program structure (10%)
        - Well-structured code
        - Correct use of assembly syntax
    - Program implementation (15%)
        - Correct implementation of the function with correct output
        - Implemented program works without error and matches the specification
        - Consistent programming style and use of comments
- Part 2 Any 2 Questions, each (35%)
    - Program presentation (5%)
        - Detailed description of the data types and data structures used
        - Source code reasonably indented, commented and readable
    - Program structure (10%)
        - Well-structured code
        - Correct use of assembly syntax
    - Program implementation (20%)
        - Correct implementation of the function with correct output
        - Implemented program works without error and matches the specification
        - Consistent programming style and use of comments
        - Evidence of the implementation of subroutines and the correct call

You will receive a % mark in each of these categories. The overall mark will be a percentage (0-100%).

How will you get feedback?

Your tutor will mark the assessment and provide you with a written feedback sheet. You can use this feedback to guide your further learning on the module.

Submission Requirements:
Your submission must include the following:

Report – Use the provided report template.
Zip File – Containing all code files, ensuring they are in working order. Marks will be awarded based on functionality.
Evaluation Criteria:
Your code will be tested against specific conditions, including but not limited to:

ATI guidelines and requirements
Few tips to consider for maximum grade:

- Use the report template provided and focus on technical overview and explanations. Use debugging tables to show how values will move from one block to other.

- For the mandatory Task 1 Part 1, please make sure you have used printing command to show output. Please check VLE files to review how it is done. Sample file is there. I understand, sometimes, storing and printing can be tricky in computer architecture but at least use the code block and you will be awarded marks, even if it doesn't print output but you have tried your best to run the print block.

- For Part 2, Task 2 requires an output of ASCII sum. Rest of the tasks are optional for output generation, as long as you state where you tried returning the subroutines in your code.

there is something called subroutine you must use it